

Supplementary evidence table

Hierarchical Taxonomic Abstraction for the Safe Handling of Novel Objects in Autonomous Driving Perception (v1, doi:10.5281/zenodo.21593472)

Accompanies the review exchange. Per-detection data: `cases.csv` (54 rows). This sheet reports aggregates with explicit denominators.

Evaluation set. SegmentMeIfYouCan Road Anomaly, 40 images, 54 detections. Detector: YOLOv8s, class-agnostic NMS, confidence ≥ 0.20 . CLIP: ViT-B/32. No training or fine-tuning.

Population split (novelty is a proxy). A detection is labelled *out-of-taxonomy* when the detector’s COCO class has no node in our taxonomy (e.g. giraffe, airplane). The detector already recognizes these objects; this is a *label-space gap*, not open-world novelty. It is distinct from genuine, scale-dependent undecidability (the “shampoo-bottle” regime), which is not tested here. Out-of-taxonomy $n = 20$; in-taxonomy $n = 34$.

Thresholds. Leaf-softmax temperature $\tau = 0.06$; commit-mass $\theta = 0.40$ (minimum subtree probability mass to descend one level); minimum best-leaf cosine $\sigma = 0.20$ (below this: immediate UNKNOWN). Safety floors: Living Being, Vehicle, Static Object.

Population	Metric	Flat (argmax)	Flat (reject@0.5)	Hierarchical
Out-of-taxonomy ($n = 20$)	safe-handled	0/20	0/20	20/20
	(abstracted / UNKNOWN)	—	—	9 / 11
	of which <i>plausibly</i> abstracted/flagged	—	—	15/20
	importance-weighted safe	0%	0%	100%
	confident wrong leaf	20/20	0/20	0/20
	dropped (missed)	0/20	20/20	0/20
In-taxonomy ($n = 34$)	useful category	16/34	—	26/34
	honest UNKNOWN	—	—	8/34
	wrong branch	—	—	0/34

Definitions. *Safe-handled* (out-of-taxonomy): the detection is neither given a specific leaf nor dropped; it is abstracted to a category at or below a safety floor, or flagged UNKNOWN. *Useful category* (in-taxonomy): outcome is not UNKNOWN, the reported node is below a floor and on the correct root-to-leaf branch. The flat baseline’s 0/20 is defined by the comparison rule (arg-max can only emit a leaf; the reject variant drops), not yet by a fully equivalent empirical open-set baseline.

Plausibility (stricter than safe-handled). *Safe-handled* only requires that a detection is not given a confident leaf and not dropped. A stricter check, *plausibly abstracted / flagged*, additionally requires the coarse category to be semantically correct: 15/20 pass (11 flagged UNKNOWN, 4 abstracted to the correct floor), while 5 abstract to an implausible branch (airplane, kite and frisbee to *Living Being*). Those are out-of-ODD classes with no correct branch in the taxonomy, which is precisely the gap a genuine open-world novelty test must target. See the `plausible` column in `cases.csv`.

Selection rule (manifest). SegmentMeIfYouCan Road Anomaly, validation split, the first 40 samples in Hugging Face streaming order (deterministic, no shuffle). The `image` column in `cases.csv` (0..39) is the stream position; each row is one detection. Per-image content hashes can be provided on request for exact identification.

Revised results statement. Across 40 images, YOLOv8s produced 54 detections; 20 were out-of-taxonomy (the predicted COCO class has no taxonomy node). All 20 satisfied the predefined fallback rule (9 abstracted, 11 UNKNOWN): *20/20 fallback-rule coverage for detected label-space-gap proxies*. This is not novelty-detection accuracy and not demonstrated driving safety. Both the population assignment and the reference class derive from the YOLO prediction, so the test shows internal behaviour relative to a proxy label, not independent semantic correctness.

Fallback breakdown (out-of-taxonomy, $n = 20$).

Category	Count
Conservative UNKNOWN (honest flag)	11
Proxy-consistent branch abstraction (correct floor)	4
Cross-branch proxy inconsistency (airplane, kite, frisbee $\rightarrow$ Living Being)	5
Detection-stage misses	unmeasured (no box GT)
Fallback coverage	20/20
Semantic-branch plausibility	15/20

Preliminary evidence-to-envelope mapping (design intent, *not* evaluated; authorization is execution-governance-owned). Perception evidence does not authorize action; it is one input to an action-specific decision.

Perception output (with residual uncertainty)	Permissible envelope
IDENTIFIED, leaf below floor, channels agree	class-specific manoeuvre
ABSTRACTED to a plausible floor	class-level caution only (decel., separation)
UNKNOWN, or cross-branch / low confidence	restricted operation / Review / Safe-Hold
cross-channel disagreement or high uncertainty	down-rank the envelope one step

Scope and caveats. (i) Novelty is a label-space proxy, not open-world novelty. (ii) Population and reference class both derive from the YOLO prediction, not independent object-level ground truth. (iii) No box-level ground truth: object-level false negatives and missed objects are outside the evaluation; the KPI scores only detected objects. (iv) Sensor degradation and broader domain shift are not evaluated. (v) The taxonomy provides bounded semantic *evidence*; authorization of any manoeuvre is a separate (execution-governance) decision.